

2. Location analysis: Analyze the location data associated with tweets to understand where particular location are most prevalent.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv('location.csv')
df.head()
```

	Unnamed: 0.1	Unnamed: 0	Text	Sentiment	Timestamp	User	Platform	Hashtags	Retweets	Likes	Country	Year	Month	Day	H
0	0	0	Enjoying a beautiful day at the park! ...	Positive	2023-01-15 12:30:00	User123	Twitter	#Nature #Park	15.0	30.0	USA	2023	1	15	
1	1	1	Traffic was terrible this morning. ...	Negative	2023-01-15 08:45:00	CommuterX	Twitter	#Traffic #Morning	5.0	10.0	Canada	2023	1	15	
2	2	2	Just finished an amazing workout! 👏 ...	Positive	2023-01-15 15:45:00	FitnessFan	Instagram	#Fitness #Workout	20.0	40.0	USA	2023	1	15	
3	3	3	Excited about the upcoming weekend	Positive	2023-01-15	AdventureX	Facebook	#Travel #Adventure	8.0	15.0	UK	2023	1	15	

Next steps: [Generate code with df](#) [View recommended plots](#) [New interactive sheet](#)

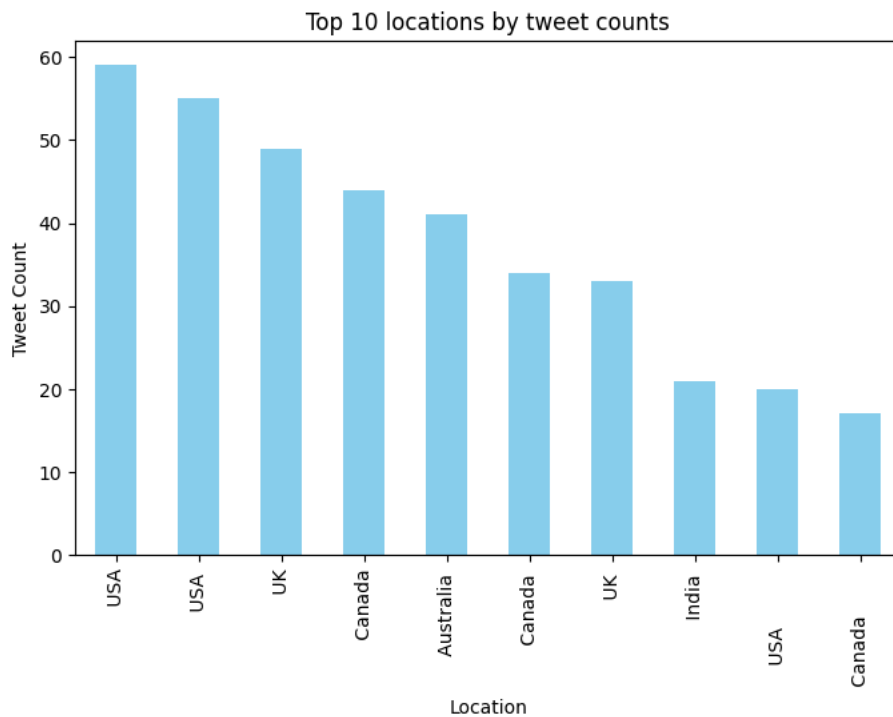
```
location_ctr = df['Country'].value_counts()
print(location_ctr)
```

```
Country
USA      59
USA      55
UK       49
Canada   44
Australia 41
..
Ireland   1
Scotland  1
Kenya     1
Jamaica   1
Thailand   1
Name: count, Length: 115, dtype: int64
```

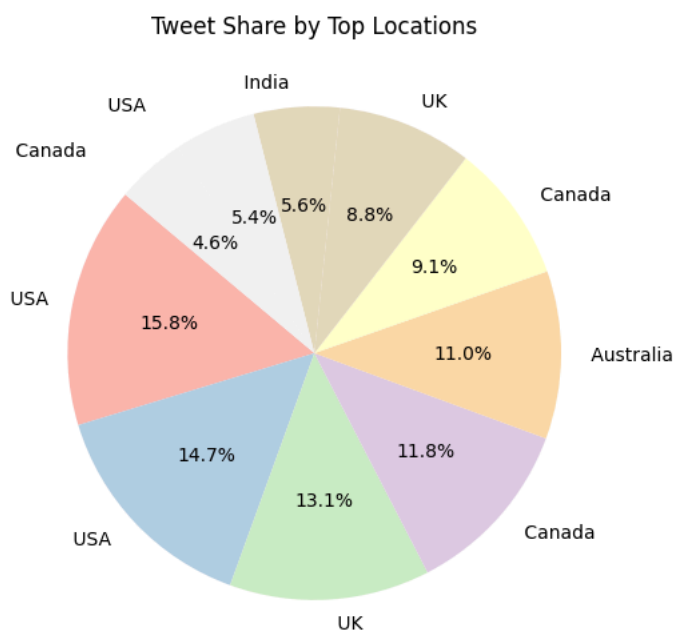
```
top_locations = location_ctr.head(10)
print(top_locations)
top_location_names = top_locations.index.tolist()
df_top = df[df['Country'].isin(top_location_names)]
```

```
Country
USA      59
USA      55
UK       49
Canada   44
Australia 41
Canada   34
UK       33
India    21
USA      20
Canada   17
Name: count, dtype: int64
```

```
top_locations.plot(kind='bar', figsize=(8,5), color='skyblue')
plt.title('Top 10 locations by tweet counts')
plt.xlabel("Location")
plt.ylabel("Tweet Count")
plt.show()
```



```
top_locations.plot(kind='pie', autopct='%1.1f%%', startangle=140, figsize=(6,6), colormap='Pastel1')
plt.title("Tweet Share by Top Locations")
plt.ylabel("")
plt.show()
```



```
sentiment_by_location = df[df['Country'].isin(top_location_names)].groupby(['Country', 'Sentiment']).size().unstack().fillna(0)

# Plot using matplotlib
sentiment_by_location.plot(kind='bar', stacked=True, figsize=(10,6), colormap='coolwarm')
plt.title("Sentiment by Top Locations")
plt.xlabel("Location")
plt.ylabel("Tweet Count")
plt.xticks(rotation=45)
plt.legend(title="Sentiment")
plt.tight_layout()
plt.show()
```

 <ipython-input-28-8ed59046b077>:10: UserWarning: Tight layout not applied. The bottom and top margins cannot be made large enough to
plt.tight_layout()



